

Cardiovascular Disease Detection from ECG and Chest X- ray

Methodology and Results — Unimodal Model Development and Combined Diagnostic Evaluation

This chapter presents the development, training, and evaluation of the two unimodal deep learning components that form the diagnostic core of this thesis: a convolutional network operating on 12-lead ECG signals, and a transfer-learning classifier operating on chest radiographs. Both models are trained and validated on the same paired patient cohort, which allows their outputs to be examined against the clinical record for the same admissions and, ultimately, alongside each other. The chapter is organised in three parts: Part One covers the ECG-based model, Part Two covers the chest X-ray-based model, and Part Three describes how the two trained models are brought together into a single diagnostic workflow.

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Part One — ECG-Based Cardiac Condition Detection

1.1 Dataset

The data used in this work comes from the **MIMIC-IV** clinical database, which contains de-identified inpatient records from Beth Israel Deaconess Medical Center, Boston. From this database, a final curated dataset was prepared containing patients who had both an ECG recording and a chest X-ray obtained during the same hospital admission. This paired structure was maintained intentionally to support the combined diagnostic view developed in Part Three.

Metric	Value
Total ECG recordings	5,926
Unique patients	5,218
Sampling frequency	500 Hz
Recording duration	10 seconds
Number of leads	12
Mean patient age	66.6 years (range: 18–99)

Each ECG is stored as a `.hea` header file (signal metadata) and a `.dat` binary file (raw 16-bit waveform). The 12 standard leads are: I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, and V6.

1.2 Target Conditions and Ground Truth

The system detects seven cardiac conditions, grouped by clinical category as displayed in the application:

Group	Condition
Rhythm	Atrial Fibrillation, Sinus Tachycardia
Conduction	Left Bundle Branch Block, Right Bundle Branch Block
Ischaemia / Infarct	ST Ischaemia, Anterior Infarct
Interval	Prolonged QT

These seven conditions were chosen because together they span the four categories a 12-lead ECG is principally read for — rhythm, conduction, ischaemia, and interval abnormality — giving the model coverage across the diagnostic breadth of the ECG rather than a narrow slice of it.

Labels were extracted from the **ECG Machine Report** column in the patient dataset summary, the automated text report generated by the ECG device at acquisition. Each condition was assigned a binary label using keyword matching against the report text:

Condition	Example Keywords
ST Ischaemia	"st elevation", "st-t changes"
Atrial Fibrillation	"atrial fibrillation", "afib"
Left Bundle Branch Block	"left bundle branch", "lbbb"
Prolonged QT	"prolonged qt", "long qt"
Anterior Infarct	"anterior infarct", "anterior mi"
Right Bundle Branch Block	"right bundle branch", "rbbb"
Sinus Tachycardia	"sinus tachycardia"

Label Prevalence

Condition	Positive Cases	Prevalence
ST Ischaemia	1,495	25.2%
Sinus Tachycardia	1,289	21.8%
Atrial Fibrillation	704	11.9%
Anterior Infarct	650	11.0%
Right Bundle Branch Block	583	9.8%
Left Bundle Branch Block	348	5.9%
Prolonged QT	340	5.7%

The dataset is **multi-label** — one ECG can carry more than one condition — and this class imbalance was directly addressed in the training design below.

1.3 Signal Preprocessing

Four steps are applied to every ECG before it enters the model, during training, validation, and inference alike:

1. **Physical unit conversion** — Raw 16-bit integer values are converted to millivolts using the gain and baseline stored in the `.hea` header.
2. **Outlier clipping** — Each lead is clipped to $\pm 5 \times \text{IQR}$ from the median, removing artefact spikes while preserving genuine high-amplitude features such as tall R waves.
3. **Z-score normalisation** — Each lead is independently centred and scaled to zero mean and unit variance, giving consistent input amplitude regardless of recording device or patient.
4. **Length standardisation** — All signals are fixed to 5,000 samples (10 seconds at 500 Hz); shorter recordings are zero-padded, longer ones truncated.

1.4 Model Architecture — ECGNet

ECGNet is a custom one-dimensional deep convolutional neural network with **23,065,634 trainable parameters** (~23 million), built from three components.

Multi-Scale Stem. The input (12 leads × 5,000 samples) is processed by three parallel convolutional branches — a short kernel (7 samples / 14 ms) for QRS morphology, a medium kernel (15 samples / 30 ms) for P- and T-wave shape, and a long kernel (31 samples / 62 ms) for ST-segment and slow baseline trends. Each branch uses stride-2 convolution; their outputs are concatenated and fused through a 1×1 convolution to 64 channels, followed by max-pooling to a (64, 1250) feature map.

Residual Stages with Squeeze-and-Excitation Attention. The network body contains **20 SE-ResBlocks** across four stages:

Stage	Channels	Blocks	Parameters
Stage 1	64	4	235,528
Stage 2	128	4	881,161
Stage 3	256	8	7,226,385
Stage 4	512	4	14,043,145

Each block applies two convolutional layers with a residual skip connection, followed by a Squeeze-and-Excitation (SE) module that learns to scale each feature channel up or down based on the global content of the signal — amplifying channels carrying relevant cardiac patterns and suppressing noisy ones, so the network can adaptively focus on different ECG features depending on what is present.

Dual-Pool Classification Head. After Stage 4, the feature sequence (512, 157) is summarised by Global Average Pooling and Global Max Pooling in parallel, concatenated to 1,024 dimensions, and passed through:

```
Linear(1024 → 512) + BatchNorm + ReLU + Dropout(50%)
Linear(512 → 256) + BatchNorm + ReLU + Dropout(30%)
Linear(256 → 7) → probability per condition (sigmoid)
```

Each output is an independent probability between 0 and 1, so multiple conditions can be flagged for the same ECG simultaneously.

1.5 Training Configuration

Dataset Split (Patient-Level)

The split is performed at the **patient level** — all ECGs from the same patient fall into the same subset — to prevent the model from memorising patient-specific morphology rather than generalisable patterns.

Split	Patients	Purpose
Training	70% (3,654)	Model weight optimisation
Validation	15% (782)	Checkpoint selection, threshold calibration
Test	15% (782)	Final held-out evaluation

Loss Function

Standard binary cross-entropy is ill-suited to imbalanced multi-label problems, since the large number of negative examples dominates the gradient. **Asymmetric Focal Loss** applies different focusing strengths to positive and negative examples:

Positive loss:

$$-(1 - p)^{1.0} \times \log(p)$$

Negative loss:

$$-(1 - \tilde{p})^{8.0} \times \log(1 - \tilde{p})$$

With a negative focusing exponent of 8.0, easy negative examples contribute almost no gradient, so the model trains mainly on the cases it is getting wrong. Label smoothing of 0.02 reduces overconfidence on noisy labels. Rarer conditions additionally receive higher loss weights (inverse square root of frequency), with further boosts for Anterior Infarct ($\times 2.5$), ST Ischaemia ($\times 1.8$), and Prolonged QT ($\times 1.8$); a weighted random sampler mirrors these priorities during batch construction.

Data Augmentation

Augmentation	Rate	Effect
Time shift (± 600 ms)	Always	Reduces overfitting
Amplitude scaling ($\times 0.75\text{--}1.25$)	Always	Gain variation
Gaussian noise ($\sigma = 0.03$)	Always	Acquisition noise
Baseline wander	55%	Electrode movement
50 Hz powerline interference	30%	Mains noise
Lead dropout (1–2 leads)	20%	Poor electrode contact
Signal masking (100–500 samples)	25%	Signal interruption
Mixup (blend two samples)	50%	Soft-label regularisation

Key Hyperparameters

Hyperparameter	Value
Optimiser	AdamW
Peak learning rate	8×10^{-4}
Weight decay	5×10^{-5}
LR schedule	OneCycleLR (10% warmup + cosine decay)
Gradient clipping	max norm = 1.0
Batch size	8
Training epochs	40
Early stopping patience	18 epochs

Stochastic Weight Averaging (SWA) was applied over the final 30% of training; averaging weights across the last epochs produces a solution in a flatter region of the loss landscape that generalises better than any single checkpoint, and the SWA model was saved as the final

checkpoint.

Training Progression

Epoch	Train Loss	Val Loss	Mean Accuracy	Mean AUC
1	0.0840	0.0589	46.4%	0.778
10	0.0545	0.0471	88.1%	0.869
20	0.0472	0.0445	88.5%	0.891
30	0.0396	0.0444	89.3%	0.905
40	0.0371	0.0472	90.1%	0.906

The best checkpoint was saved at epoch 34 (validation AUC 0.906); the final SWA model reached **mean AUC 0.907**.

Threshold Calibration

A per-condition threshold was selected on the validation set by searching thresholds from 0.05 to 0.95 and choosing the value that maximises F1 subject to a minimum precision of 45%, rather than using a single fixed threshold for every condition regardless of its prevalence.

Condition	Threshold
ST Ischaemia	0.57
Atrial Fibrillation	0.60
Left Bundle Branch Block	0.64
Prolonged QT	0.60
Anterior Infarct	0.58
Right Bundle Branch Block	0.62
Sinus Tachycardia	0.62

1.6 Results

(Acc = Accuracy; Prec = Precision; Rec = Recall; AFib = Atrial Fibrillation; RBBB = Right Bundle Branch Block; LBBB = Left Bundle Branch Block)

Condition	AUC	F1	Acc	Prec	Rec
Sinus Tachycardia	0.994	93.0%	97.2%	90.0%	96.2%
RBBB	0.994	90.2%	97.9%	86.0%	94.7%
LBBB	0.984	89.3%	98.9%	86.2%	92.6%
AFib	0.984	86.2%	96.3%	89.0%	83.7%
Prolonged QT	0.968	71.2%	95.7%	60.4%	86.7%
Anterior Infarct	0.934	68.8%	90.8%	69.1%	68.6%
ST Ischaemia	0.886	70.4%	81.5%	63.0%	79.7%

Sinus Tachycardia (AUC 0.994) is the strongest result: an elevated heart rate sustained across the full recording is a simple, repeating temporal pattern, and the model detects it with 96.2% recall and a negative predictive value of 99.1%.

Right Bundle Branch Block (AUC 0.994) produces a stereotyped morphological signature — a secondary R wave (rSR') in V1/V2 and a wide slurred S wave in I, V5, V6 — consistent enough across patients to give 94.7% recall.

Left Bundle Branch Block (AUC 0.984) is one of the least prevalent conditions (5.9%) yet is still detected with 92.6% recall and 86.2% precision, reflecting the effect of the class-weighting and augmentation strategy on low-frequency labels; its broad monophasic R wave and QS pattern in V1 form a distinctive template.

Atrial Fibrillation (AUC 0.984) was detected in 83.7% of confirmed cases at a precision of 89.0%, meaning false alarms remained rare relative to correct detections. Absent P waves and an irregularly irregular ventricular response are consistently encoded across the recording; the missed cases likely correspond to AF with a well-controlled ventricular

rate, where the irregularity is subtler.

Prolonged QT (AUC 0.968) shows high recall (86.7%) but lower precision (60.4%). QT prolongation is a continuous measurement rather than a binary state, and some borderline cases visible in the waveform are not explicitly called out in the machine report; the strong AUC confirms the underlying probability ranking is sound.

Anterior Infarct (AUC 0.934) shows balanced errors on both sides. The label spans three distinct presentations — acute ST elevation, subacute T-wave inversion, and old pathological Q waves — which the model must learn jointly, and its anterior-lead pattern overlaps with both ST Ischaemia and LBBB.

ST Ischaemia (AUC 0.886) is the most difficult condition, driven by high prevalence (25.2%, the most common finding in the cohort) and a broad keyword definition that also captures non-ischaemic causes of ST-T change. The AUC is still clinically meaningful — true cases are ranked above non-cases 89% of the time — but the output is best read as a screening signal rather than a definitive diagnosis.

1.7 Diagnostic Application

A desktop application, **ECG_main.py**, was built on the trained model using Python, tkinter, and matplotlib, and runs entirely offline. A user selects a `.hea` ECG file, clicks Analyse, and the application renders the 12-lead waveform in the standard 3×4 clinical layout with a Lead II rhythm strip below, looks up the patient's chart-recorded diagnosis, and displays a results table with the **Actual Result** (from the chart) alongside the **AI Prediction** and its confidence for each of the seven conditions. Each row is colour-coded — agreement, a missed chart-confirmed condition, or a condition the AI flags that the chart does not record — so that a reviewer can open any record and see immediately where the model and the clinical documentation agree. Inference runs in a background thread to keep the interface responsive.

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Part Two — Chest X-ray-Based Cardiac Condition Detection

2.1 Dataset

The chest radiographs used in this part of the work are drawn from the same MIMIC-IV cohort described in Part One — patients for whom both a 12-lead ECG and a chest X-ray were obtained during the same hospital admission. This shared-cohort design is what allows the two independently trained models to be exercised on the same patient in Part Three.

Metric	Value
Chest radiographs available	6,318
Unique patients	5,218
Image format	Frontal chest radiograph (AP or PA), PNG
Input resolution	224 × 224, single channel

The great majority of the source studies were acquired in the anteroposterior (AP) projection using portable bedside equipment (83% of studies with a recorded projection), with the remainder acquired postero-anteriorly (PA) in an upright radiography suite — consistent with the acutely admitted, often bed-bound population described for the ECG cohort in Part One.

2.2 Target Conditions and Ground Truth

- **Cardiomegaly** — enlarged cardiac silhouette, a direct radiographic marker of chronic cardiac strain or failure.
- **Pulmonary Oedema** — fluid accumulation in the lung interstitium/alveoli, a hallmark of acute decompensated heart failure.

- **Pleural Effusion** — fluid collecting around the lungs, frequently secondary to elevated cardiac filling pressures.

These three findings were chosen because each is a recognised radiographic correlate of cardiac dysfunction, keeping the chest X-ray branch deliberately aligned with the cardiovascular focus of this thesis rather than attempting general-purpose chest radiograph interpretation.

Ground truth labels were taken from the structured radiology labels associated with each study — the standard labelling scheme used for MIMIC-CXR reports, of which Cardiomegaly, Edema, and Pleural Effusion form the subset relevant here. Each label takes one of three states: present, absent, or uncertain/not mentioned; studies with an uncertain or unmentioned status for a given condition were excluded from the loss and evaluation for that condition only, rather than being forced into a positive or negative class.

Label Prevalence

Condition	Prevalence
Cardiomegaly	68.9%
Pleural Effusion	61.1%
Pulmonary Oedema	41.6%

All three conditions are markedly more prevalent here than in a general chest radiograph population, which is expected: every patient in this cohort was also under ECG monitoring during the same admission, weighting the sample toward patients under active cardiac investigation.

2.3 Image Preprocessing

Each radiograph is converted to grayscale, resized to 224×224 pixels — the input resolution expected by the pretrained backbone — and its pixel intensities rescaled from the 8-bit $[0, 255]$ range to the $[-1024, 1024]$ range used during the backbone's original pretraining, so that the input distribution seen at inference matches the distribution the backbone was trained on.

2.4 Model Architecture — Transfer Learning with DenseNet-121

Rather than training a convolutional network from randomly initialised weights on a comparatively modest cardiac-paired cohort, this branch uses **transfer learning**: a DenseNet-121 backbone, pretrained on a large combination of public thoracic radiograph datasets (NIH ChestX-ray14, PadChest, CheXpert, and MIMIC-CXR, via the `torchxrayvision` library), supplies the visual feature representation, and only a compact classification head is trained on top of it for the three target conditions. A backbone exposed to several hundred thousand chest radiographs across multiple institutions has already learned a general-purpose visual vocabulary for the thorax — lung fields, cardiac silhouette, costophrenic angles, pleural space — so a much smaller cohort is sufficient to teach the network which combinations of those learned features correspond to the target conditions.

- **Backbone** — DenseNet-121 (`densenet121-res224-all` weights), frozen.
- **Feature extraction** — final dense block, ReLU, then global average pooling to a 1,024-dimensional feature vector.
- **Classification head** — `Linear(1024→256)`, BatchNorm, GELU, Dropout(0.5), `Linear(256→3)`.
- **Output** — sigmoid probability per condition.

Parameter Group	Count	Trainable
Backbone (DenseNet-121)	6,966,034	No
Classification head	263,683	Yes
Total	7,229,717	3.6% trainable

Because the backbone weights are held fixed, the effective learning problem is a linear-probe classification task over a 1,024-dimensional feature space rather than full end-to-end image classification, considerably reducing the labelled data and compute needed to reach a stable operating point.

2.5 Training Configuration

Dataset Split (Patient-Level)

As in Part One, the split is performed at the patient level, using the same 70/15/15 partition of the shared cohort, so that no patient's data appears in more than one subset for either modality.

Split	Patients	Purpose
Training	70% (3,654)	Classification head optimisation
Validation	15% (782)	Checkpoint selection, threshold calibration
Test	15% (782)	Final held-out evaluation

Two-Phase Training Pipeline

Training proceeds in two phases. First, every radiograph is passed once through the frozen backbone to extract its 1,024-dimensional feature vector; because the backbone never changes during training, this extraction happens once rather than being repeated every epoch. Second, the lightweight classification head is trained directly on these fixed feature vectors, which makes each epoch fast since no image convolution is required during head training.

Loss Function

A **masked binary cross-entropy** loss is used, evaluated independently per condition; where a condition's ground truth is uncertain or unrecorded for a given study, that condition's contribution to the loss is masked out for that study rather than scored against a guessed label. A per-condition positive class weight is additionally applied — the square root of the ratio of positive to negative training examples, clipped to `[1.0, 4.0]`:

Condition	Positive Weight
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Cardiomegaly	1.74
Pleural Effusion	1.35
Pulmonary Oedema	1.00

Key Hyperparameters

Hyperparameter	Value
Optimiser	Adam (classification head parameters only)
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
LR schedule	Cosine annealing
Batch size	256
Training epochs	150

Training Progression

Epoch	Train Loss	Val Loss	Val AUC
1	0.717	0.651	0.822
3	0.557	0.497	0.828
50	0.443	0.462	0.821
150	0.414	0.464	0.818

Because the backbone is frozen and only a small head is being fitted to a fixed, already-informative feature space, the head converges within the first few epochs, with the best validation AUC (0.828) reached at epoch 3. The remainder of the run confirmed the operating point was stable, and the checkpoint at the best validation AUC was carried forward to threshold calibration and testing.

Threshold Calibration

Per-condition decision thresholds were calibrated on the validation set using **Youden's J statistic** (sensitivity + specificity – 1), searched over 0.05 to 0.95 in steps of 0.005 — the criterion that best balances sensitivity against specificity, appropriate for a screening tool where both missed findings and false alarms carry a clinical cost.

Condition	Threshold
Cardiomegaly	0.595
Pulmonary Oedema	0.555
Pleural Effusion	0.625

2.6 Results

(*Acc* = Accuracy; *Prec* = Precision; *Sens* = Sensitivity/Recall; *Spec* = Specificity)

Condition	AUC	F1	Acc	Prec	Sens	Spec
Pleural Effusion	0.829	82.6%	77.0%	87.9%	77.9%	75.0%
Pulmonary Oedema	0.826	74.8%	75.6%	82.1%	68.8%	83.2%
Cardiomegaly	0.822	80.0%	72.0%	93.2%	70.1%	79.7%
Macro average	0.826	79.1%	74.9%	87.7%	72.3%	79.3%

Pleural Effusion (AUC 0.829) is the strongest of the three findings. Costophrenic-angle blunting and a meniscus-shaped fluid line remain visible across a wide range of severities and projections, which transfers well from the backbone's general thoracic representation; precision of 87.9% means positive flags are usually correct.

Pulmonary Oedema (AUC 0.826) is the most balanced result, with sensitivity and specificity in a similar range. Oedema is a severity spectrum rather than a binary finding — from subtle interstitial change to frank alveolar shadowing — which is a natural source of the residual

error on both sides of the decision boundary.

Cardiomegaly (AUC 0.822) has the highest precision of the three (93.2%) but the lowest sensitivity (70.1%): when the model flags cardiomegaly it is usually correct, but it misses a meaningful share of true cases. Cardiomegaly is conventionally read from the cardiothoracic ratio on an upright PA film; the great majority of the source studies here are portable AP films, which introduce magnification and rotation that make borderline cases harder to call consistently from a single projection.

Across all three conditions, the AUC values (0.822–0.829) are appreciably lower than the strongest ECG conditions in Part One but broadly comparable to the ECG model's harder conditions such as ST Ischaemia (AUC 0.886). This is consistent with the underlying task: rhythm and conduction abnormalities produce highly stereotyped, repeating waveform signatures, whereas radiographic findings are graded along a continuous severity spectrum and are additionally affected by projection, patient positioning, and technique. The results are nonetheless clinically meaningful as a screening signal: an AUC around 0.82–0.83 means the model correctly ranks a positive case above a negative case in roughly four out of five random comparisons for each condition.

2.7 Diagnostic Application

A second desktop application, `xray_main.py`, mirrors the structure of the ECG tool in Section 1.7. A user selects a chest X-ray image, and the application displays the radiograph alongside a results table showing the **Clinical Record** (Confirmed / Negative / Not in Report) next to the **AI Result** and predicted probability for each of the three target conditions, with each row colour-coded to distinguish agreement, a missed recorded finding, a finding flagged without a matching record, and a finding flagged for a study with no recorded outcome at all. As with the ECG tool, this layout turns the aggregate statistics in Section 2.6 into something a reviewer can inspect case by case.

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Part Three — Combining ECG and Chest X-ray Analysis

3.1 Per-Modality Comparison Against the Clinical Record

The ECG model in Part One and the chest X-ray model in Part Two were trained entirely independently — separate architectures, separate loss functions, separate training pipelines — and at no point does either model see the other modality's data during training. What connects them is the underlying cohort: the dataset was built specifically from patients with both an ECG and a chest X-ray recorded during the same hospital admission, so the two trained models can be pointed at the same patient and their outputs considered together.

For each modality individually, the diagnostic applications in Sections 1.7 and 2.7 compare the model's prediction against the outcome recorded in the clinical chart, condition by condition, for whichever record is loaded. The two comparisons are kept separate rather than blended into a single score, because the ECG record and the radiology record are independent sources of ground truth generated by different clinical processes — collapsing them would obscure exactly where each model does, and does not, agree with its own documentation. This is what makes it possible to state, as in Sections 1.6 and 2.6, precisely how reliable each model is against its own ground truth.

3.2 Combined Diagnostic Application

A third desktop application, **main.py**, brings the two validated models into a single full-screen interface. A user loads one ECG recording and one chest X-ray for the same patient through two dedicated upload panels, then triggers a single **Analyze** action that runs both models and returns a consolidated results screen: the twelve-lead ECG waveform with its seven conditions grouped by clinical category on one side, and

the chest radiograph with its three conditions on the other, each finding shown with a detected/not-detected indicator and its confidence percentage. Unlike the two single-modality tools in Sections 1.7 and 2.7, this view does not compare the predictions against the clinical record — it presents the two models' outputs only, giving a single consolidated screening summary for a patient from one combined action rather than a case-by-case audit against ground truth.

3.3 Independent, Non-AI Interpretability Verification

Beyond producing a probability per condition, the combined application addresses a limitation common to deep learning-based diagnostic tools: a model can report high confidence in a prediction without any externally checkable indication of *why*. A widely used approach to this problem is gradient-based saliency, which visualises the region of the input that most influenced the network's own decision. Such methods characterise only the model's internal attention: they confirm where the network was looking, not whether that region is medically appropriate for the condition being reported, and two networks can reach the same correct prediction by attending to different, even spurious, regions without a saliency map revealing the difference. Saliency-based explanation was therefore deliberately not used in this work.

Instead, every prediction surfaced by the application is paired with a second, independent verification step, computed after inference and without access to the network's weights, activations, or gradients. This step uses established, deterministic signal-processing and image-processing techniques, applied only to conditions the model has already flagged as positive, to locate the region of the input that independently corresponds to that condition. The model therefore answers *whether* a condition is present, and this second, separate process answers *where* the supporting evidence is located — two independently obtained answers rather than one process explaining itself.

3.3.1 ECG Waveform-Region Verification

For the ECG signal, waveform delineation is performed using the `neurokit2` library, which identifies the onset, peak, and offset of the P

wave, the QRS complex, and the T wave for every beat in the twelve-lead recording using classical peak-detection and wavelet-based delineation, the same landmarks used in manual ECG interpretation, rather than a learned model. For each condition the network reports as positive, the corresponding physiologically expected segment is located from these landmarks and shaded on the twelve-lead display in a colour specific to that condition:

Condition	Verified Region
ST Ischaemia	ST segment (QRS offset → T onset)
Anterior Infarct	Initial third of the QRS complex (Q-wave portion)
Left / Right Bundle Branch Block	Full QRS complex
Prolonged QT	QRS onset → T offset
Atrial Fibrillation	Expected P-wave window
Sinus Tachycardia	Each full beat (QRS onset → next QRS onset, the R–R interval)

Only conditions the model has flagged as positive are shaded, so the display never implies evidence for a condition the network itself did not report. Where the ten-second recording ends before delineation of the final beat can complete — the algorithm requires the onset of the following beat to close out the previous one with confidence — a short extension of signal beyond the displayed window is read from the source file where available, purely to supply this lookahead context; no boundary derived from that extension is drawn outside the original ten-second display.

3.3.2 Chest X-ray Geometric Verification

For the chest radiograph, verification is performed directly on the pixel data using Otsu's thresholding method and connected-component geometry, independently of the classification head. Otsu's method selects the intensity threshold t that maximises the between-class variance of the pixels split into a foreground and background class at

that threshold:

$$\sigma^2_b(t) = \omega_0(t) \cdot \omega_1(t) \cdot [\mu_0(t) - \mu_1(t)]^2$$

where ω_0 , ω_1 are the class probabilities and μ_0 , μ_1 the class means for a given threshold t ; the threshold that maximises $\sigma^2_b(t)$ is selected automatically for each image rather than fixed in advance.

This threshold is applied in two stages. First, it separates the patient's body from the image background, and the largest connected bright region is retained and filled to give a single thoracic-cavity mask. Second, within the central-lower portion of that mask — the region bounded by the lungs and above the diaphragm, where the cardiac silhouette sits — a second, local Otsu threshold separates the denser cardiac silhouette from the surrounding aerated lung field.

Cardiothoracic Ratio. When Cardiomegaly is reported positive, the maximum transverse width of the segmented cardiac silhouette, W_{heart} , and the width of the thoracic cavity at that same vertical level, W_{thorax} , are measured directly from the two masks, and the Cardiothoracic Ratio is computed as:

$$CTR = \frac{W_{heart}}{W_{thorax}}$$

which is the same measurement used in conventional radiographic reporting, where $CTR > 0.50$ is the standard threshold for an enlarged cardiac silhouette. The measured value is displayed on the image together with the two reference lines used to obtain it, so the finding is expressed as an interpretable clinical quantity rather than a heatmap. Because this measurement is obtained independently of the classification head, it is not guaranteed to agree with the network's prediction; where the two disagree — the network reports Cardiomegaly positive but the measured CTR falls at or below 0.50 — the application displays this disagreement explicitly on the image rather than suppressing it, since a transparent record of disagreement between two independently obtained results is more informative than a system that only ever appears to agree with itself.

Lung-Field and Costophrenic-Angle Regions. For Pulmonary Oedema, the segmented lung fields — excluding the central mediastinal column — are shaded as the expected region of diffuse fluid accumulation. For Pleural Effusion, the lower lateral portion of the thoracic mask on each side, corresponding to the costophrenic angle, is shaded as the expected region of fluid collection.

Left/Right Laterality. For both of these findings, the mean pixel intensity is computed separately within the left- and right-side region, since denser fluid or consolidation attenuates more radiation and appears brighter than normally aerated lung. Writing $\bar{I}_L$ and $\bar{I}_R$ for the mean intensity on the left and right side respectively, the relative asymmetry between the two sides is:

$$\Delta = \frac{|\bar{I}_L - \bar{I}_R|}{\max(\bar{I}_L, \bar{I}_R)}$$

If Δ exceeds a fixed threshold of 0.12, the denser side is reported as predominantly affected and shown at full visual emphasis, while the other side is retained at reduced emphasis for context; otherwise the finding is reported as bilateral and both sides are shown at equal emphasis. This produces a side-specific reading rather than shading both lungs uniformly regardless of which side the image actually shows as affected.

3.4 Fairness Evaluation — Methodology and Results

Deep learning diagnostic systems trained on real clinical data can inherit disparities in performance across patient subgroups even when no demographic attribute is used as a model input, if the training distribution is itself imbalanced across those subgroups. To assess this directly rather than assume it away, the application includes a built-in fairness evaluation that compares each model's discrimination performance between male and female patients on a held-out test set.

3.4.1 Metric and Statistical Procedure

For a given condition and patient group, discrimination performance is summarised by the Area Under the Receiver Operating Characteristic Curve (AUC), equivalent to the probability that the model ranks a randomly chosen positive case above a randomly chosen negative case:

$$\text{AUC} = P(s(x^+) > s(x^-))$$

for a positive example x^+ , a negative example x^- , and model score $s(\cdot)$, estimated on the group-specific subset of the held-out test set.

A point AUC estimate alone does not indicate whether an observed difference between two groups reflects a real disparity or sampling noise, particularly where one group has few positive cases. Two further statistics were therefore computed for every condition, using non-parametric bootstrap resampling with 1,000 resamples:

- A **95% confidence interval** for each group's AUC, obtained by resampling that group's cases with replacement, recomputing the AUC on each resample, and taking the 2.5th and 97.5th percentiles of the resulting distribution.
- A **two-sided significance test** for the male/female gap, obtained by independently resampling each group with replacement, recomputing $\text{AUC}(\text{male}) - \text{AUC}(\text{female})$ on every resample to build an empirical distribution of the gap, and reporting

$$p = 2 \cdot \min(P(\Delta_{\text{AUC}} \leq 0), P(\Delta_{\text{AUC}} \geq 0))$$

against that distribution — a bootstrap-based alternative to a DeLong test, used here so that no assumption of a particular sampling distribution for the AUC difference is required. A condition is additionally flagged as a **small-sample result**, to be interpreted cautiously rather than taken as a confirmed finding, whenever either group has fewer than 25 positive cases in the test set.

3.4.2 Held-Out Test Set Verification

Fairness evaluation is only meaningful if the patients used for it were genuinely not seen during model training. For the chest X-ray model, the

same patient-level 70/15/15 split used during training was reconstructed directly from the original training script and applied to the current dataset, and its validity was confirmed by observing realistic, non-perfect AUC on the resulting held-out group, consistent with the values reported in Section 2.6.

For the ECG model, an initial reconstruction of the split using an equivalent procedure produced implausible results: every condition scored a near-perfect AUC on both the nominal "held-out" group and a comparison group, which is not attainable by a genuinely unseen evaluation and indicated that the reconstructed split did not correspond to the patients the model was actually trained on. The original patient-level split was traced to its source data and recovered directly, and evaluation was repeated against the correct held-out group; the resulting AUC values (0.786–0.994 across all nine ECG conditions, 0.878–0.994 across the seven reported in Section 1.6) are consistent with the validation-time performance already reported in Section 1.6, confirming the recovered split as genuinely held-out. This verification step is reported here because a fairness evaluation is only as trustworthy as the independence of the data it is computed on, and an unverified split would risk reporting fairness metrics that are silently optimistic.

3.4.3 Results

The evaluation covers all seven ECG conditions and all three chest X-ray conditions against the recovered held-out test set (529 male / 419 female ECG cases; 167–247 male and 145–204 female chest X-ray cases per condition, reflecting the ground-truth availability described in Section 2.2).

ECG conditions (529 male / 419 female cases):

Condition	AUC – M	AUC – F	Gap	p	Flag
ST Ischaemia	0.856	0.907	–0.051	0.032	Significant
Atrial Fibrillation	0.982	0.986	–0.003	0.702	—
Left Bundle Branch	0.955	0.994	–0.039	0.190	Small sample

Prolonged QT	0.985	0.944	+0.041	0.090	—
Anterior Infarct	0.931	0.927	+0.003	0.858	—
Right Bundle Branch	0.989	0.991	−0.003	0.618	—
Sinus Tachycardia	0.995	0.994	+0.001	0.666	—

Chest X-ray conditions (167–247 male / 145–204 female cases):

Condition	AUC – M	AUC – F	Gap	p	Flag
Cardiomegaly	0.836	0.902	−0.066	0.168	—
Pulmonary Oedema	0.880	0.861	+0.019	0.624	—
Pleural Effusion	0.913	0.867	+0.047	0.146	—

(M = Male, F = Female; AUC values are point estimates from the held-out test set; a dash in the Flag column means no significant gap and an adequate sample size.)

9 of the 10 evaluated conditions show no statistically significant gender gap ($p > 0.05$). The single exception is **ST Ischaemia** (Male AUC 0.856, 95% CI [0.823, 0.889]; Female AUC 0.907, 95% CI [0.874, 0.935]; $p = 0.032$), where the model discriminates positive from negative cases more reliably for female patients than for male patients on this held-out set. This is reported directly rather than adjusted away: ST Ischaemia is also the ECG condition with the lowest overall AUC in Section 1.6 (0.886), driven by its broad, high-prevalence keyword definition, and the present result suggests that residual difficulty is not evenly distributed between male and female patients. All other conditions in both modalities show overlapping confidence intervals and non-significant p-values, indicating no detectable gender-based disparity in discrimination performance at the sample sizes available.

[INSERT FIGURE 3.1 HERE] — In Word: place the cursor on this line, delete this bracketed note, then **Insert** → **Pictures** → **This Device** and select `fairness_auc_chart.png` (same folder as this

file). Caption: *"Figure 3.1 — Gender fairness evaluation: AUC per condition for male and female patients on the held-out test set, with 95% bootstrap confidence intervals. The asterisk marks the one condition (ST Ischaemia) with a statistically significant gender gap."*

3.5 Comparative Reliability of the Two Modalities

Taken together, the two branches show a consistent pattern: conditions with a stereotyped, repeating signature — the rhythm and conduction abnormalities read from the ECG — are detected with very high reliability (AUC up to 0.994), while conditions graded along a continuous severity spectrum and sensitive to acquisition technique — the radiographic findings, and to a lesser extent ST-segment and interval abnormalities on the ECG — are detected with moderate but still clinically useful reliability (AUC 0.82–0.94). This pattern indicates that the two modalities carry complementary rather than redundant diagnostic information: each is strongest on a different class of cardiac abnormality, which is precisely why bringing both into the same combined interface, as in Section 3.2, gives a fuller screening picture than either modality alone.

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4. Novelty and Contribution

This work departs from the majority of existing cardiac AI studies, which typically develop a single-modality model on a generic public dataset, in several concrete respects.

A deliberately paired, single-modality-pair cohort. Rather than drawing the ECG and chest X-ray branches from two unrelated public datasets, both models are trained and evaluated on data curated from the same underlying MIMIC-IV cohort, restricted specifically to patients who have both an ECG and a chest X-ray from the same hospital admission. This shared-patient structure is what makes it possible, for

the first time in this project, to run both trained models on one patient's data and view both results together rather than as two disconnected reports.

A purpose-built ECG architecture rather than an adapted off-the-shelf network. ECGNet was designed specifically for this task: a multi-scale stem processes three temporal resolutions in parallel, tuned respectively to QRS morphology, P/T-wave shape, and ST-segment trends, feeding into 20 Squeeze-and-Excitation residual blocks and a dual-pool (average and max) classification head. This is a bespoke combination of components chosen for the specific waveform characteristics of the seven target conditions, rather than a generic convolutional or recurrent network borrowed unchanged from another domain.

Clinically grouped, independently calibrated conditions. Both branches treat their target conditions as a clinically organised set — rhythm, conduction, ischaemia/infarct, and interval for the ECG; three heart-failure-related radiographic findings for the chest X-ray — rather than a flat, undifferentiated label list, and every condition in both branches receives its own independently calibrated decision threshold instead of a single fixed cut-off applied uniformly.

Domain-pretrained transfer learning for the imaging branch. The chest X-ray branch is built on a backbone pretrained directly on chest radiographs drawn from several public thoracic-imaging datasets, rather than the generic natural-image pretraining (e.g. ImageNet) used by many transfer-learning baselines, so the visual representation available to the classification head is already specific to the thoracic domain before any cardiac-focused training begins.

Reviewer-facing, case-level comparison against the clinical record. Beyond the aggregate AUC/F1 tables reported in Sections 1.6 and 2.6, both single-modality diagnostic tools display the AI's prediction directly alongside the actual outcome recorded in the clinical chart for the specific record being viewed, condition by condition. This lets a reviewer judge model reliability on an individual patient basis, not only as a summary statistic.

A working combined interface across two independently trained

modalities. The application described in Section 3.2 demonstrates that the two independently developed branches can already be operated together on shared patient data through a single action, producing one consolidated cardiovascular screening summary from two independent physiological signals — a capability the two single-modality tools alone do not provide.

Independent, non-AI interpretability verification. Rather than explaining a prediction through the network's own internal attention, every detected condition is paired with a second, deterministic verification step — neurokit2 waveform delineation for the ECG, and Otsu-threshold image geometry, including an explicitly computed Cardiothoracic Ratio, for the chest X-ray (Section 3.3) — computed without access to the model's weights or gradients, so that the displayed evidence can be checked independently of the network that produced the prediction, and any disagreement between the two is shown rather than hidden.

Built-in, statistically grounded fairness auditing. Gender-based performance disparity is evaluated as a first-class part of the application itself rather than a separate offline analysis, using bootstrap confidence intervals and a significance test for every condition in both modalities (Section 3.4), with small-sample results explicitly flagged rather than reported with false confidence.

A verified, leakage-free evaluation methodology. The held-out test split underlying the fairness evaluation was independently verified rather than assumed correct: an initial split reconstruction was rejected after producing implausible near-perfect results, and the genuine original patient-level split was traced to its source and recovered before any fairness result was reported (Section 3.4.2), demonstrating the evaluation methodology itself withstands scrutiny rather than merely producing convenient numbers.

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5. Overall Reliability and Conclusion

Across both branches, the models achieve macro-average AUC values of 0.907 (ECG, seven conditions) and 0.826 (chest X-ray, three conditions). The strongest individual conditions in each modality — Sinus Tachycardia and Right Bundle Branch Block on the ECG side, Pleural Effusion on the radiograph side — reach a level of discrimination that would be considered strong by conventional diagnostic-accuracy standards, and the weaker conditions in each branch remain useful as a screening signal even where they fall short of a diagnostic standard. In every case the residual error can be traced to a specific, explainable property of the underlying clinical task — label ambiguity in free-text ECG reports, or severity-grading and projection variability in chest radiographs — rather than to an unexplained model failure.

The fairness evaluation in Section 3.4 extends this reliability picture across patient subgroups: 9 of the 10 evaluated conditions show no statistically significant difference in discrimination performance between male and female patients, and the one exception — ST Ischaemia, already the weakest-performing ECG condition overall — is reported explicitly rather than concealed, together with the verification steps that establish the evaluation itself as trustworthy. Combined with the independent, non-AI interpretability layer in Section 3.3, which allows every prediction to be checked against a separately computed clinical measurement or waveform region rather than the model's own self-reported attention, the system offers not only two complementary diagnostic signals but an explicit account of where each prediction comes from and how consistently it performs across patients.

Both unimodal models are reliable enough, individually, to support the combined diagnostic workflow in Part Three, and together — with independent interpretability verification and a statistically grounded fairness audit built into the same application — they demonstrate that cardiovascular screening drawn from two independent physiological signals can be delivered through a single, practical, and accountable diagnostic interface built entirely from paired, real clinical data.

Data: MIMIC-IV and the associated MIMIC-CXR chest radiograph collection (PhysioNet). All patient identifiers are de-identified. Research use only.